Green Paper 01 — Moral Biology

Ethics as capacity in bodies, relations, institutions, and living worlds

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Steward: Lars A. Engberg · Spiralweb Stewardship Association

Nothing in nature blooms all year.
The moon wanes.
The bear sleeps.
The soil lays fallow.

Rest is not a reward — it is a right.
Productivity is not proof of worth.
Burnout is not a badge of honor — it is a boundary breached.

There are moments when a body knows before a sentence does.

A breath becomes shallow. Attention contracts. A task that was possible yesterday becomes impossible today. A person still knows what ought to be done, yet cannot find the capacity to do it without injury. A relationship still contains care, yet no longer has enough room to carry another demand. An institution continues to speak the language of responsibility while exhausting the people through whom responsibility must become real.

The body is not a private error message.

It may be signalling something about a life, a relation, a workplace, an economy, or a world whose demands have exceeded what can be carried. The signal is not automatically a verdict. It may be partial, frightened, displaced, or mistaken. But it is part of the reality any adequate ethics must learn to hear.

This paper begins there.

Abstract

Moral Biology names a simple proposition: ethics is not only a question of values, principles, or intentions. It is also a question of capacity.

People and institutions may know what ought to be done and still be unable to carry it. Under sustained strain, attention narrows, time horizons shorten, trust becomes expensive, repair recedes, and moral responsibility is displaced onto the very people with the least remaining room. These are not merely private failures. They are patterned interactions between bodies, relationships, institutions, economies, habitats, and ecological limits.

Moral Biology does not derive morality from biology, reduce ethics to physiology, or treat bodily feeling as infallible truth. It asks a different question: what living conditions make ethical perception, judgment, consent, cooperation, responsibility, and repair possible — and what conditions quietly make them impossible?

The paper develops ethics as a distributed capacity across body, relation, institution, and habitat. It treats overload as morally consequential without making it morally exculpatory; embodied signals as full but fallible knowledge; care as dependent on viable conditions; and planetary limits as part of the material jurisdiction within which moral life takes place. It concludes with corrigible wagers and questions for practice.

A note on the term

Moral Biology emerged independently during the artistic and analytical inquiry AbunDance in 2025. At the time, Lars A. Engberg was not aware of earlier uses of the phrase. It arose through sustained attention to nervous-system regulation, embodiment, relational life, institutional pressure, and lived experience: the recurring observation that people may know what ought to be done while lacking the bodily, relational, or material capacity to carry it.

The phrase has appeared previously in other contexts, and this paper makes no claim to having coined it. Its meaning here is specific. Moral Biology does not seek to derive morality from biology, explain ethics through genes or brain mechanisms, or replace normative judgment with physiology. It asks how living conditions shape the capacity for ethical agency — across bodies, relationships, institutions, habitats, and planetary constraints.

The term did not begin as a theory imposed upon life.

It began as something repeatedly encountered:

the body tightening before language;

care collapsing under overload;

responsibility becoming heavier than a person or institution could carry;

and ethical possibility returning when breath, rhythm, trust, and sufficient ground were restored.

The concept is therefore offered neither as a new scientific discipline nor as a proprietary name. It is a working lens within a wider, correctable inquiry.

1. A quiet proposition

People can know what ought to be done — and still be unable to do it.

This does not abolish responsibility. It makes responsibility more exact.

Most ethical languages begin with judgment: What is right? What is just? What do we owe one another? These questions remain indispensable. But they are not self-executing. To perceive a situation with some openness, to tolerate ambiguity, to resist a coercive demand, to remain present in conflict, to imagine a longer horizon, to admit error, or to repair harm all require capacities that can be supported or depleted.

Ethics is not only what we believe.

It is what we can carry.

A person deprived of sleep, safety, time, food, dignity, or meaningful choice does not encounter a moral situation from the same ground as a person who is rested and protected. A worker asked to compensate indefinitely for an institution's understaffing does not simply face a personal challenge of resilience. A community required to absorb repeated ecological or economic shocks is not failing morally when its attention contracts toward immediate survival. A decision system that demands nuance while rewarding speed produces some of the very failures it later attributes to individuals.

Capacity does not decide what is right. It conditions whether right action remains reachable.

That distinction is the beginning of Moral Biology.

2. Biology is not destiny, and feeling is not law

The word *biology* carries risks. It has often been used to naturalise hierarchy, excuse violence, reduce social life to competition, or present historically produced conditions as inevitable. Moral Biology must refuse that inheritance explicitly.

It does not claim:

- that moral norms can be read directly from nature;
- that brain states determine moral worth;
- that stress excuses harm;
- that embodied intuition outranks evidence or the testimony of others;
- that institutions are literally organisms;
- that what survives is therefore good;
- or that the capacity to act determines whether a person deserves dignity.

Biology here means that moral life is enacted by living beings with thresholds, dependencies, histories, vulnerabilities, and powers of recovery. We breathe, tire, attach, grieve, orient, defend, learn, and sometimes lose access to capacities we still value. We do this within relations and material environments that enter the body continuously through labour, food, water, danger, shelter, language, care, pollution, noise, expectation, and time.

The fact that a signal is embodied does not make it pure. Bodies carry trauma, prejudice, memory, desire, social conditioning, illness, and fatigue. A tightening may indicate danger; it may also reflect an old defence activated in a new situation. A feeling is real as an event even when its interpretation is wrong.

The discipline is therefore neither obedience to the body nor mastery over it. It is attentive inquiry.

We can distinguish:

> I noticed.
> I infer.
> I know.
> I remain uncertain.

Embodied knowledge becomes more trustworthy through repetition, comparison, material consequence, dialogue, and willingness to revise. The body is part of the instrument. Like any instrument, it must be cared for, situated, and checked.

3. The body is not a private error message

Modern institutions frequently privatise the signals they produce.

Exhaustion becomes poor self-management. Anxiety becomes insufficient resilience. Moral distress becomes lack of professionalism. Withdrawal becomes disengagement. Illness becomes absence. Burnout becomes an individual event recorded after the system has already consumed the warning.

This framing protects institutional innocence. If the signal belongs only to the person, the organisation need not ask what pattern of demand, fragmentation, contradiction, exposure, or unacknowledged labour helped produce it.

Moral Biology proposes a wider reading.

A body may register institutional conditions before the institution has language for them. Dread before a recurring meeting, chronic vigilance in an opaque process, numbness under unresolvable responsibility, or fatigue that lifts when a coercive demand is removed may be data about more than individual psychology. These experiences do not prove a diagnosis of the whole. They are situated signals that deserve a place in collective inquiry.

To say that the body is not a private error message is not to make private experience public property. No institution is entitled to extract interior life in the name of learning. People retain the right to silence, privacy, refusal, and exit. What must become public is not every intimate detail, but the possibility that repeated bodily strain indicates a patterned condition for which governance bears responsibility.

The ethical movement is:

from *What is wrong with this person?*

to *What is this person encountering, and what does the pattern ask us to examine?*

Sometimes the answer will still include individual responsibility, harmful conduct, illness, error, or a need for specialist care. Moral Biology is not an all-purpose explanation. Its contribution is to prevent the living signal from being dismissed before the wider field has been considered.

4. Body · relation · institution · habitat

The earlier version of this paper named three connected levels: body, relation, and institution. The inquiry has since made a fourth explicit: habitat.

These are not separate containers. They are connected conditions of action.

Body

The body sets limits and makes response possible. Attention, energy, sleep, sensory load, illness, nourishment, pain, safety, and recovery influence what a person can perceive and carry. No symbol, strategy, or institution can responsibly require unlimited capacity from a finite organism.

Your body is the altar.
Your presence is the prayer.

Read publicly, these lines do not ask for worship or assent. They restore consequence to the place where abstraction is felt. A policy eventually enters someone's breathing, muscles, sleep, workday, water, food, or home.

Relation

Capacity is co-regulated. Trust can widen a person's ability to remain present; coercion can narrow it. A boundary spoken early may preserve a relation that silent endurance would destroy. Rupture is not the opposite of relationship. The absence of repair is.

Relationships also distribute hidden work. One person remembers, anticipates, translates, reassures, hosts, absorbs conflict, or remains continuously reachable. When this labour is unnamed, a relation may appear viable only because one nervous system is financing it.

Moral Biology asks not only whether people care for one another, but whether the relation allows care to continue without capture or depletion.

Institution

Institutions shape moral outcomes through pace, staffing, incentives, categories, information flows, default priorities, decision rights, and what they make visible or leave unrecorded. Harm does not require malicious intention. It can emerge through ordinary fragmentation: no one holds the whole consequence, each procedure appears defensible, and responsibility dissolves between departments, contracts, dashboards, and deadlines.

An institution has ethical capacity when it can receive difficult information without punishing the messenger; distinguish urgency from chronic acceleration; retain memory; name who makes the final decision; pause a harmful movement; repair after failure; and change a rule when reality contradicts it.

Habitat

Bodies and institutions exist within air, water, soil, climate, buildings, infrastructures, species relations, and places with their own histories. Habitat is not scenery around moral life. It is part of its material ground.

Soil is not dirt.
It is memory.
It is a living library
of everything we've ever buried —
and everything still willing to grow.

The line is poetic, but its orientation is concrete. Soil holds residues of use, damage, repair, organic matter, water, cultivation, and neglect. A place carries consequences forward. Ethical agency that ignores habitat mistakes a temporary human frame for the whole field of consequence.

5. Moral overload

Moral overload occurs when responsibility exceeds the capacity, authority, time, or support available to carry it.

It is common in care work, public administration, teaching, activism, ecological stewardship, families, and precarious communities. A person is expected to notice everything, respond well, remain kind, document correctly, prevent harm, absorb urgency, and compensate for missing resources — often while having little power to alter the conditions generating the demand.

The result is not necessarily indifference. It may be the collapse of reachable responsibility.

Under sustained overload:

- attention becomes selective and threat-oriented;
- the near term displaces the long term;
- procedural completion substitutes for judgment;
- people protect themselves through numbness, withdrawal, or rigidity;
- conflict becomes harder to metabolise;
- mistakes become more difficult to admit;
- care becomes heroic and therefore fragile;
- responsibility travels downward while authority remains elsewhere.

This pattern can create a cruel moral loop. The system depletes capacity, interprets the resulting contraction as personal failure, responds with more monitoring or exhortation, and thereby increases the depletion.

The alternative is not lower ethical expectation. It is an insistence that expectation be joined to carrying conditions.

If a school depends on one exhausted teacher, a community garden on invisible unpaid labour, a local ecological effort on a steward who cannot rest, or a public service on permanent emergency, the apparent success contains its own degradation.

Burnout is not proof that a person cared too much.

It may be evidence that rotation stopped, boundaries were ignored, or an institution financed its continuity through hidden human depletion.

6. Care requires capacity

Care is often described as virtue, intention, or disposition. It is also an arrangement of time, attention, resources, skill, trust, and recovery.

Nothing in nature blooms all year.

This is not a universal biological law. It is a cultural correction to systems that treat continuous output as proof of value. Living processes contain pulses, seasons, pauses, dormancy, repair, and uneven emergence. Human work cannot simply imitate nature, but neither can it remain viable while denying that bodies have rhythms.

Rest is not a reward — it is a right.

To call rest a right is to move it from private indulgence into institutional design. A right that cannot be exercised without punishment is only rhetoric. Rest requires staffing, income, boundaries, redundancy, realistic scope, and cultures in which stepping back does not mean disappearing from worth.

Care becomes viable when systems make room for:

- a legitimate no;
- sufficient time to understand;
- responsibility proportionate to authority;
- recovery after exposure;
- rotation of visible and invisible burdens;
- material support without ownership or capture;
- repair without compulsory reconciliation;
- incompletion without humiliation;
- and an unknown status when reality cannot yet be read honestly.

This is where Moral Biology becomes institutional rather than therapeutic. The answer to structurally produced depletion cannot be an endless programme of individual self-regulation. Breathing practices, rest, friendship, movement, and clinical care can matter deeply. They cannot repair an institution that continues to manufacture the same injury.

Regulation is not adaptation to everything.

Sometimes regulation is the capacity to stop.

7. Three streams that must not cancel one another

The later Spiralweb architecture developed a practical discipline that sharpens the argument of this paper. A living field can be read through three distinct streams:

1. **Land / Ecology** — is the place regenerating?
2. **Steward Viability** — can the people carry the work without hidden depletion?
3. **Governance / Commons** — are roles, consent, decisions, and responsibility sufficiently clear?

The streams remain separate. A green reading in one cannot cancel a red reading in another.

This matters because many systems aggregate unlike realities into one success score. Ecological improvement can conceal burnout. Efficient administration can conceal degraded soil. Strong community enthusiasm can conceal unclear consent or untraceable decisions. Money raised can conceal pressure placed on the people whose lives are used to justify the appeal.

Moral Biology supplies the ethical reason for non-compensation: life-supporting capacities are related, but they are not interchangeable.

If steward viability goes red, ecological ambition must pause. This is not because human comfort always outranks other life. It is because regeneration financed by concealed depletion reproduces the pattern it claims to correct.

The pause is not failure.

It is the system becoming capable of telling the truth about itself.

8. Responsibility without shame

An account of capacity can be misused as an excuse: *I was stressed; therefore I am not responsible.* That is not the claim.

Responsibility and condition must be held together.

A person may have acted under severe strain and still have caused harm. An institution may be overloaded and still owe repair. A community may be vulnerable and still reproduce exclusion. Understanding the conditions of action should make response more intelligent, not make consequence disappear.

Shame often collapses identity and action: *I did harm* becomes *I am irredeemably harmful*. This can produce hiding, denial, counterattack, or paralysis. Responsibility asks for something more demanding: the capacity to remain present to consequence, acknowledge one's part, protect those affected, make restitution where possible, and change the conditions that would reproduce the harm.

In the Bloom Treaty, no one is thrown away.
But nothing is hidden either.

This does not promise reintegration in every relation. Those harmed retain boundaries and do not owe access, forgiveness, or renewed trust. The line describes a social horizon: refusing both disposability and concealment.

Moral Biology therefore seeks responsibility without shame, but never responsibility without consequence.

9. Planetary constraint enters the room

The body is finite. So is the planet.

This resemblance must not be made simplistic: Earth is not a human organism, and planetary boundaries are not bodily feelings writ large. But both interrupt fantasies of unlimited substitution. There are conditions beyond which continued pressure changes what remains possible.

Ethics under planetary constraint must carry delayed effects, irreversible losses, unequal exposure, and the knowledge that those contributing least to degradation often meet its consequences first. It must also resist the temptation to place the entire weight of planetary responsibility on individual nervous systems.

No person can feel the whole Earth continuously and remain functional. Planetary care therefore requires distributed sensing, protected local knowledge, institutions able to receive slow signals, and economic arrangements that allocate capacity toward places and people carrying real consequence.

The planetary does not become ethical by becoming abstract.

It becomes ethical when distant consequence is allowed to alter present action without erasing the authority of people in actual places.

Moral Biology joins a long horizon to a consequence-close scale. A watershed, workplace, neighbourhood, food forest, public institution, household, or ten-square-metre observation plot can become a place where the larger condition is encountered without pretending that one place represents the whole.

10. From signal to correction

A signal becomes ethically useful when it can enter a correction process.

The movement is simple:

1. **Notice** — something changes in a body, relation, institution, or habitat.
2. **Locate** — who or what is affected, and where is the consequence carried?
3. **Distinguish** — what was observed, what is inferred, and what remains unknown?
4. **Compare** — is the signal repeated, shared, contradicted, or materially supported?
5. **Pause when necessary** — do not demand certainty before stopping possible harm.
6. **Decide accountably** — a named human or legitimate body holds the final decision.
7. **Act narrowly** — make the smallest sufficient movement rather than expanding the intervention.
8. **Return** — observe what changed, including displaced burdens and unintended effects.
9. **Correct** — revise the action, rule, interpretation, or framework without punishing truthful feedback.

AI may assist articulation, comparison, translation, and pattern recognition within this movement. It cannot feel consequence, hold moral responsibility, certify readiness, or replace situated judgment. The final accountable impulse remains human and locatable.

Documentation also has limits. Not every feeling should be recorded. Not every interior signal belongs in a ledger. Public memory should preserve decisions, claims, consent, material movement, and relevant correction without converting a person's inner life into institutional data.

The aim is not total legibility.

It is sufficient shared memory for responsibility to remain possible.

11. What Moral Biology changes

If the proposition of this paper is provisionally accepted, several ordinary questions change.

Instead of asking only:

> Did people comply?

we also ask:

> Could they meaningfully choose?

Instead of:

> Was the project successful?

we ask:

> What did the success require from land, bodies, relationships, and future capacity?

Instead of:

> Why is this person resistant?

we ask:

> What boundary, history, or consequence may the resistance be protecting?

Instead of:

> How do we motivate more care?

we ask:

> What conditions would allow care to continue without heroism?

Instead of:

> How do we scale the solution?

we ask:

> What must become more distributed so that scale does not make the centre heavier?

Instead of treating ethics as an additional demand placed upon exhausted systems, Moral Biology asks whether the system itself protects the capacities through which ethical life becomes possible.

12. Corrigible wagers

These are working wagers, not settled findings:

1. **Ethical judgment has carrying conditions.** Attention, time, safety, material sufficiency, relational trust, and institutional support affect whether people and groups can act on values they sincerely hold.
2. **Repeated bodily strain may contain institutional information.** It should neither be privatised automatically nor treated as self-interpreting truth.
3. **Care becomes fragile when its costs remain invisible.** Naming and distributing those costs can increase rather than diminish care.
4. **A system that cannot pause cannot correct itself.** The right to stop is part of ethical infrastructure.
5. **Unlike life-supporting capacities should not be collapsed into one score.** Ecological gain, steward viability, and governance integrity must remain separately legible.
6. **Responsibility becomes more exact when condition and consequence are held together.** Explanation need not become exoneration; accountability need not become shame.
7. **Planetary responsibility requires distributed capacity.** No centre can sense or carry the whole; truthful scale begins in protected local relation and returns through shared learning.
8. **Technologies can extend perception without receiving moral authority.** Assistance must remain correctable, and the final accountable impulse must remain human.

Each wager may be narrowed, revised, or abandoned when practice contradicts it.

13. Questions for practice

- Where does responsibility currently exceed authority or carrying capacity?
- Which bodily or relational signals are repeatedly privatised as individual weakness?
- What hidden labour makes the present arrangement appear viable?
- Who is standing permanently at the exposed edge?
- What would have to change for rest, refusal, or uncertainty to become legitimate?
- Which success measure allows one healthy stream to hide another that is failing?
- Can affected people halt the movement without punishment?
- Where does documentation support memory, and where does it become extraction?
- Who holds the last accountable decision when tools, procedures, and institutions interact?
- What is the smallest correction that could restore room to breathe?

Closing

Moral Biology begins before moral certainty.

It begins where life meets demand.

Where a body tightens.

Where care becomes too heavy to carry alone.

Where an institution discovers that its values have no material home.

Where soil, water, workers, families, stewards, and future generations receive consequences that were absent from the decision.

It asks us to listen without surrendering judgment, to take limits seriously without worshipping them, and to make responsibility possible without pretending that good intentions can substitute for capacity.

To lie down is to remember you are held.
To be still is to rejoin the intelligence of the dark.

There is no promise here that rest will solve what is structural, or that every signal will become clear.

There is only a constitutional orientation:

No ethics that destroys the capacity for ethical life can remain sufficient.

And no system should call itself caring while treating the living beings who carry it as infinitely available.

Textual provenance and working lineage

The italic passages are drawn from *Gaia Bloom Constitution* (AbunDance, 2025). Both that poetic source field and this later Spiralweb return were formed through sustained human–AI inquiry. Lars A. Engberg holds the lived inquiry and serves as steward of the text: selecting, correcting, releasing, and remaining answerable for it.

The argument also grows through the later Spiralweb inquiry: *The Correction Loop*, *Penguin Dashboard: Legibility as Governance*, *SRIP — The Steward's Journey*, *Regenerative Reciprocity*, *Knowing From the Ground*, *Rule of Life*, *Silent Degradation*, *The Spiral of Hypotheses*, and *A Decent Economy under the Jurisdiction of Life*.

Selected wider intellectual companions include:

- Archer, M. S. (2000). *Being Human: The Problem of Agency*. Cambridge University Press.
- Bhaskar, R. (1979). *The Possibility of Naturalism*. Harvester Press.
- Churchland, P. S. (2011). *Braintrust: What Neuroscience Tells Us about Morality*. Princeton University Press.
- McEwen, B. S. (1998). Protective and damaging effects of stress mediators. *The New England Journal of Medicine, 338*(3), 171–179.
- Ostrom, E. (1990). *Governing the Commons*. Cambridge University Press.
- Tronto, J. C. (1993). *Moral Boundaries: A Political Argument for an Ethic of Care*. Routledge.

These references form a working lineage rather than a claim of comprehensive review. Empirical claims require verification in the context where they are used.

Read with

- Green Paper 02 — *Regulation: Body · Relation · Institution*
- Green Paper 03 — *Viability and the Conditions for Care*
- Green Paper 12 — *Penguin Economics: Rotation as Care*
- Green Paper 13 — *Meatball Before Symbol: The First Constitution*
- Green Paper 16 — *Nervous-System Love*
- Report 02 — *The Correction Loop*
- Report 04 — *Penguin Dashboard: Legibility as Governance*
- *Knowing From the Ground*
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